

cAMpanion: An Ambient Light Box Connecting Humans to Pet Dogs

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ABSTRACT

Pet owners form strong emotional bonding with their pet dogs, but sometimes they have to be separated from their pet dogs due to varying reasons resulting in negative feelings such as distress or anxiety. Researchers and designers in HCI have explored a variety of approaches to resolve this issue, including designing pet camera, pet chat system, as well as devices for playful remote interaction and haptic simulation. Informed by prior work on facilitating intimacy between distant couples and family members, we present the design and implementation of *cAMpanion* to promote distant owner's sense of connectedness to their pet dogs when separating from each other. *cAMpanion* displays three statuses of pet dogs via different light colors in real-time enabled by sensor modules installed in pet dog's living environment. We share lessons learned from a pilot study examining potential issues during sensor deployment as well as participants' feedback toward *cAMpanion*.

CCS CONCEPTS

• ; • Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

KEYWORDS

ambient display, human-pet interaction, awareness, connectedness

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1 MOTIVATION AND BACKGROUND

Pet dogs play an important role in human life and provide owners emotional support and companionship. Previous research states that there is a strong bond and attachment between owner and their pet dog [12] – people treat dogs not only as pets but also as family members [1]. Owners regard dogs as an attachment figure [13] and would feel lonely or depressed when separating from their dogs [13]. Existing research on designing remote pet interaction uses a variety of approaches to facilitate connectedness between humans

and animals, including pet cameras, wearable devices, and devices for playful technology (e.g. play fetch) and haptic simulation. As owners treat their pets as family members, we reviewed the designs for mediating the sense of connectedness between distant families and couples to inform the design of *cAMpanion* (the combination of ambient & companion). This line of research has suggested that ambient display is often used to mediate the sense of connectedness of distant relationship, and can create a sense of co-presence to increase the sense of connectedness in a distant relationship. It tends to achieve this via presenting the current states of remote ones in the forms of light, auditory or olfactory feedback in the user's background environment in a less intrusive way [34]. Although ambient display is widely used in human-human interaction, it is rarely used in designs aiming to facilitate connectedness between owners and their pets.

This paper introduces the design and implementation of an ambient display to facilitate the sense of connectedness of owners who do not live with their pet dogs for a long time. We review existing research and designs on both human-human and human-pet connectedness, and design *cAMpanion* following the four characteristics of ambient information system [24]. This paper starts from reviewing designs on facilitating human-pet bonding and ambient displays enhancing awareness and connectedness. It then illustrates our design rationale and the implementation of *cAMpanion*, followed by sharing lessons learned from the pilot study consisting of sensor trial installation sessions at the dog side and interviews with distant owners. We hope to contribute to the research area of human-animal interaction by proposing a design solution that is less intrusive while enhancing connectedness of remote owners to their pets.

2 RELATED WORK

2.1 Designs facilitating human-pet bonding

Researchers use multiple approaches to enhance human-pet bonding, including pet cameras for monitoring [10, 11, 25], pet video chat system [8, 27], wearable devices for tracking pet's physical conditions or wellbeing [7, 17, 21, 33], devices providing playful interactions [10, 19, 25], and devices simulating remote touching of pets [11, 15]. Pet cameras allow owners to watch the video streams of pet dogs through webpages or mobile applications such as *Furbo* [31], *Petcube* [23], *Rover@Home* [25], *Lonelydog@home* [10], and *Nene* [11]. Pet video chat systems enable owners to see and chat with their pets remotely through Skype [8, 27]. These systems have two limitations. First, it is difficult for owners to see what they desire to see as pets are not always close to the cameras [8]. Second, it is a difficulty for dogs to understand and respond to verbal

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cues from the systems [19, 27], making them feel confused or even anxious [22].

Wearable devices such as *Fitbark* [7], *CompanionViz* [21], and *Whistle* [33] are non-invasive collars that enable owners to track the physical conditions of dogs via data visualizations in mobile application. Wearable devices are beneficial for owners to gain more insights into dogs' health conditions, allowing them to adjust the amounts of exercise and rest time their dogs need. Though these wearable devices help deepen owners' bonding with their dogs, [14] cautions that owners may over-rely on the quantified data, and fail to focus on pets' actual behaviors.

Devices providing playful interactions such as playing fetch and tossing treats allow owners to interact with their pets remotely [10, 19, 25, 31]. For example, *PAWSABILITIES* [19] enables owners to remotely control the tennis ball releasing system when their dogs are lying on the bed, a sign indicating dogs feel bored. These devices attempt to eliminate loneliness of pets, but research is limited in examining how pets perceive these interactions and whether these interactions did enhance connectedness. As for devices simulating remote haptic sensation, they mediate the feeling of touch between human and pet. Lee et al. design a system to transmit the haptic feedback to the remote chicken by activating the vibrotactile actuators inside the jacket worn by the chicken whenever humans touch the chicken doll at the office [15]. These designs enable novel interaction between pet and their owners, yet it remains a challenge to capture or even understand the animal's reactions toward the sense of touch.

2.2 Awareness and ambient display

The concept of awareness was first proposed in the Computer Supported Cooperative Work (CSCW) field, defined as "an understanding of the activities of others, which provides a context for your own activity" [6]. Systems supporting awareness create lightweight forms of interaction that allow users to monitor something of interest without causing unnecessary interference or efforts [28]. Based on prior research, awareness is an effective and commonly used strategy in mediating intimacy over distance [9, 16]. It is frequently used to facilitate the sense of connectedness for remote relationships between humans, but rarely between humans and animals. As the challenge remains to adopt an animal-centric awareness strategy due to the technical limitations of knowing and capturing the awareness of animals, our design uses an owner-centric awareness strategy to promote the sense of connectedness of distant owners toward their dogs by increasing their awareness of their dogs.

The concept of ambient display originates from the concept of "calm technology" [32]. Differed from technologies such as television or email that occupy users' central attention, calm technology is characterized by being able to move easily between users' central and peripheral attention. It usually stays at users' peripheral attention without disturbing their tasks, and only moves to their central attention when necessary. For example, Weiser and Brown hung a plastic string on the ceiling of the office [32]. The greater the network traffic is in the office, the greater the string swings. In most cases, this information only stays at the peripheral attention. If the users want to know the network traffic of the office, they can look at the string without interfering with the current

task. Similar to the characteristics of "calm technology", ambient information systems tend to display important yet non-crucial information, being able to move between the periphery and the center of one's attention, display subtle changes to reflect updates in information (none-distracting), and being aesthetically pleasing and environmentally appropriate [24].

2.3 Ambient display for facilitating connectedness

Ambient displays are widely used in designing for mediating connectedness in distant relationships. Wisneski et al. [34] stated that ambient displays can be used to create a non-intrusive and permanent connection between love ones, generating a sense of community through sharing awareness. Through long-term field experiments, researchers have also assessed the effectiveness of ambient displays on promoting the connectedness between people living apart [3, 5, 26, 28], and the result showed that ambient display enables users to create a sense of continuous awareness and connectedness to the other person. *The Digital Family Portrait (DFP)* is designed for supporting "peace of mind" for extended family members by providing awareness of the daily activities of elderly parents who are living apart [28]. The system deploys multiple sensors in elderly parents' homes to capture their in-home movement and calculate their activity levels. *The DFP* is displayed on the extended family members' side. *The DFP* is a photo frame displaying elderly parents' photos, and the frame displays 28 butterfly icons representing the activity level for the past 27 days along with the current day. In similar a vein, *AugmentedMirror Presence Display* [5] is a mirror displaying a loved one's instant messenger's presence information by colors on its frame. The frame glows red when a user's loved one is offline, green when online, and blue when idle. Researchers found that participants showed a higher sense of connectedness to the loved one during the period of using *AugmentedMirror Presence Display* through field experiment.

These two examples along with prior work [2, 3, 29] suggest that researchers often choose daily objects to serve as ambient display mediums to display information as it can better fit into users' living environment. However, these works mainly focus on the connectedness between humans such as remote families or couples, and there remain little ambient displays focusing on human's remote connectedness to the dog. To our knowledge, there are currently two ambient display designs for pet dogs [19, 30], yet their goals do not lie in fostering connectedness between humans and pets. Specifically, *DogPulse* [30] targets at supporting the coordination of dog walking job among family members, and *PAWSABILITIES* [19] displays only one dog behavior status and their experiment mainly focuses on the input and output methods on the canine side.

3 PROTOTYPE DESIGN AND IMPLEMENTATION

3.1 Design rationale of *cAMpanion*

This paper aims to present the design and implementation of an ambient display that helps enhance the connectedness between remote owners and their pet dogs. As mentioned in the Related Work section, existing designs and research on ambient displays

focus on facilitating connectedness for remote couples and remote family members. These displays tend to present information in an implicit way, integrate information into users' daily environment unobtrusively, and increase users' awareness of others to create a sense of connectedness. For instance, *The Digital Family Portrait* [28] displays an aging parent's daily activity level by a photo frame and *AugmentedMirror Presence Display* [5] presents instant messenger's status of others by a mirror. Informed by these designs and the four ambient information system characteristics proposed by Pousman and Stasko [24], *cAMpanion* uses light box as the medium to display feedback on pet dog's activities in real-time. The light box not only fits easily in the daily home environment but also delivers visual information that is easy and fast for humans to perceive and process.

Considering it remains challenging to assess how dogs perceive information from the device and send information to their owners based on prior research [18], we focus on helping increase the feeling of connectedness of distant pet owners through the *cAMpanion* ambient display. Dogs do not directly interact with the device, yet we ensure sensors are designed and installed in a non-disturbing way to pet dogs to capture their activities in a natural context. In fact, existing designs mostly adopt this approach to capture dog's activities, and that technological solutions are more limited in allowing dogs to express their physical and emotional status [19, 27]. Based on the daily routine of dogs and our initial interviews of 10 remote dog owners, we decide to present three statuses of dogs - sleeping, frequently staying in an area at home, and going out. Compared to existing pet camera or wearable trackers [7, 10, 11, 21, 23, 25, 31, 33], *cAMpanion* is low-cost and relatively easier to maintain. Additionally, research is still needed to confirm the data accuracy of dog wearable trackers, and investigate whether dogs feel comfortable wearing these trackers, and whether owners would feel overloaded by the tracking data through push notifications on the phone.

Informed by prior research [4], we use three different colors of light to present these statuses from low to high activity level - blue for sleeping, green for frequently staying at a particular area at home, and red for going out. [4] found that using three distinct colors (blue, green, and red) to present activity levels from low to high allows users to process information on activity levels more intuitively, compared to using only one color with varying degrees of brightness to display different statuses. In addition to these three colors, the light box will display white light when dogs are not in the above three statuses.

3.2 Design implementation of *cAMpanion*

Our ambient display system consists of two parts. One contains a set of sensors on the dog's side used to collect the current status and location of dogs while the other is the ambient light box on the owner's side. Once the sensors receive data, they will transmit the data to the light box through ThingSpeak [20], an Internet of Things platform, and the light box will display a particular color according to a dogs' status in real-time. The whole process takes less than one minute. Concerning the safety of the lithium battery, we use a wall socket to power the ambient light box and sensors. We provide implementation details in the following sub-sections.

3.2.1 Sensors at the dog side. There are three types of sensors on the dog side. We use the force sensor module to capture a dog's sleeping behavior, and the infrared sensor module to detect if a dog stays at his or her frequently stayed area at home. The going-out report button module is used to capture whenever dogs go out. We use the NodeMcu Lua WIFI V3 ESP8266 board to connect these sensors to Wi-Fi to transmit data to ThingSpeak in real-time. To prevent dogs from chewing or playing the sensors or causing harm, all the sensor parts were placed in a hard box and installed firmly in areas that are unreachable by the dog. Figure 1 illustrates what the real-time activity data looks like on the ThingSpeak platform.

Sensor module capturing dog's sleeping behavior. The force sensor module (Figure 2a) is placed underneath the dog's bed to determine whether the dog is sleeping on the bed by measuring changes in force. We use SparkFun square force sensitive resistor sensor to measure force once a minute. Since each dog has a different weight, the procedure is for users to measure the force value of each dog during the device setup stage and adjust the threshold value accordingly. The sensor sends "1" to ThingSpeak whenever the measured value is greater than the threshold, representing the dog is on the bed. Otherwise, it sends "0" to ThingSpeak, representing the dog is not on the bed. Please note that we use the term bed in general to indicating where a dog sleeps at a home - some dogs would sleep on the sofa while some would sleep on pet beds or other locations. We have tested the stability and functionality of the force sensor on top of different materials (e.g., ceramic/wooden floor, sofa surface) during our trial sensor installation session.

Sensor module capturing dog's frequently staying area. The infrared sensor module (Figure 2b) is installed in an area a dog stays most frequently to detect its presence in that particular area by sensing the distance between the dog and the sensor once a minute. If the distance value is less than the threshold, the sensor sends "1" to ThingSpeak, representing the dog is there. If it is greater than the threshold value, the sensor sends "0" to ThingSpeak, representing the dog isn't there. Each dog has different frequently staying area, such as under the sofa or next to the window. We have installed infrared sensor at different location (e.g., on the window, on the wall) to make sure the functionality and stability during our trial sensor installation session.

Sensor module reporting dogs going out. The going-out report button module (Figure 2c) consists of one button, one LED light bulb, and one NodeMcu Lua WIFI V3 ESP8266 board. The button will be pressed by a family member whenever they walk their dog. Once the button is pressed, the LED light bulb glows, and the ESP8266 board sends "1" to ThingSpeak, representing the dog is going out for a walk. After a period of time (depending on the length of time the family members usually walk the dog), the LED light bulb will turn off, and the ESP8266 board sends "0" to ThingSpeak, representing the dog is at home. Regarding the time length of the dog's walk, the procedure is for the users to enter in the system in advance during the set-up stage by asking participants' family members. The going-out report button module is placed near the door to make the family member easy to press the button before walking the dog.

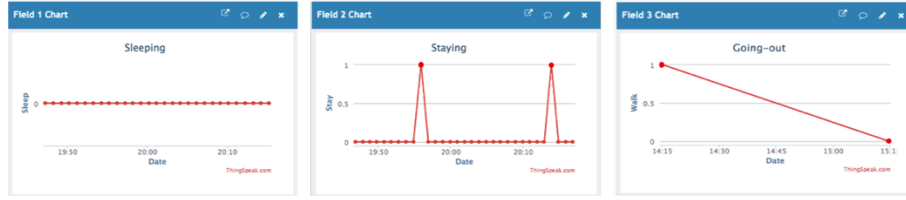


Figure 1: Real-time activity data on ThingSpeak

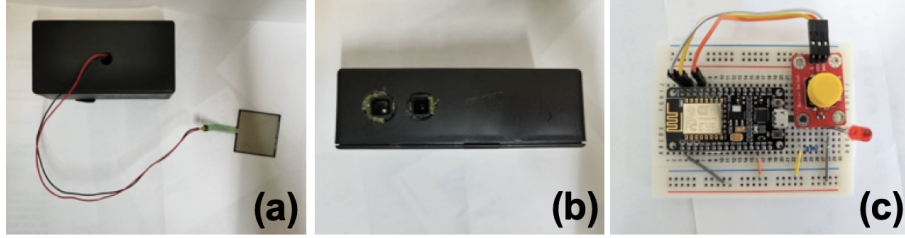


Figure 2: Sensor modules capturing dog's (a) sleeping behavior, (b) frequently staying area, and (c) going out status.

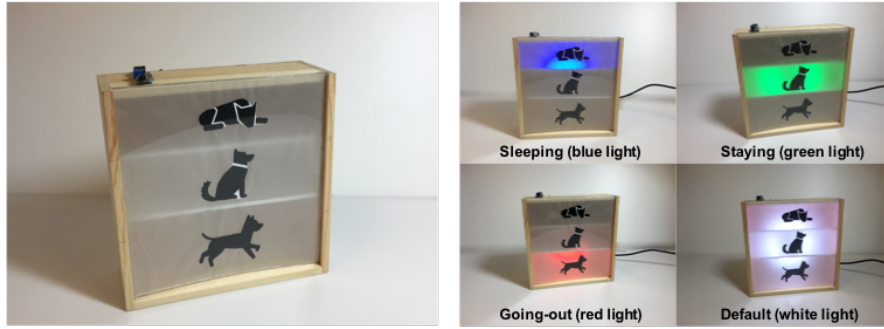


Figure 3: Ambient light box of cAMpanion

3.2.2 Ambient light box at distant-owner's side. The cAMpanion ambient display light box consists of a NodeMcu Lua WIFI V3 ESP8266 board, a NeoPixel ring, an infrared obstacle avoidance module, a wood box (7.08*7.08*2.76 inches), an acrylic board, three dog icons representing three activity statuses printed in paper (Figure 3). The paper dog icons representing three statuses (sleeping, staying at a frequent area, and going out) are pasted to the acrylic board, serving as the front side of the light box. In terms of information display, the light box receives the sensors' data from ThingSpeak through Wi-Fi enabled by the NodeMcu Lua WIFI V3 ESP8266 module in real-time. Once receiving the data, the light box displays the color via the NeoPixel ring according to the dog's status. Additionally, informed by prior designs using haptic simulation to induce a sense of togetherness with their pets, we include a breathing light feature to simulate the feeling of dog breathing by their side through dimming and brightening the light. When the user's hand reaches the sensor above the light box, the light box will present the effect of breathing light. The light box simulates the breathing of the pet dog by manipulating the dimming frequency of light. We expected this feature can create a sense of co-presence.

4 INITIAL USER FEEDBACK

We conducted a pilot study using our prototype with three dog owners who currently live away from their pet dogs, and do not see their dogs for more than five days a week. Participants are two females and one male, and age between 20 and 25 years old. The pilot study has two parts. For the first part, we installed sensors at the dog's side to see whether issues may arise during sensors deployment. For the second part, we showed participants the ambient light box to gather feedback toward the design through a semi-structured interview. Interview questions include (1) remote owners' lifestyles, and how they usually bond with their pets when living remotely, (2) reactions to the physical attributes of the ambient display light box (size, material-wood, lights, icons), and (3) their reactions when seeing 3 types of lights.

4.1 Trial sensor installation session at dog's side

Overall, there was no big issue in the trial sensor installation session (Figure 4). All sensors detected the dogs' behaviors correctly, and the data was sent to ThingSpeak in real-time without problems.



Figure 4: Sensors installed at dog's side (a) infrared sensor module, (b) force sensor module, (c) going-out report module

However, there were two things that need attention. First, since all sensors were powered through sockets via USB cables directly, we had to consider the distance between sockets and the locations where sensors were placed. We need to reconsider sensor installation location at the field site when they were too far from sockets. For instance, in most cases, pet dogs tend to stay at more than one particular areas at home, so it is important to explore multiple frequently staying areas to pick the one that is closest to the socket location. Second, although we originally expected that the force sensor module was used for capturing dog's sleeping behavior and the infrared sensor module was for capturing dogs' frequently staying area, these actually vary by dog's living habits. For instance, participant 3's dog usually sleeps on the floor without a dog bed, so it is better to detect sleeping behavior using the infrared sensor module rather than the force sensor module.

4.2 Reactions to ambient light box at distant-owner's side

Two participants we interviewed reacted positively toward the light colors. P1 mentioned the blue light gave him the feeling of relaxing, suitable to represent the dog was resting. Blue light made him think about not only the dog's status but also his family members' status ("If I see the blue light, I will wonder if my dog is sleeping because my family members are out or just because my dog is bored."). P3 thought that the white light fitted in her living environment well as it was similar to the lighting of her bedroom. P2 preferred yellow over red when showing the dog's going out status, as red light seems a warning sign for her. Moreover, two participants (P2 and P3) mentioned the brightness of the light is a bit too high, which could potentially cause the feeling of dizziness if seeing it frequently in their rooms.

As for the breathing light feature, participants 2 and 3 found this interesting. However, participants did not show strong interest in using it in their real life, as they were aware that there was no output on the dog side and their dogs did not know they touched the light box. In terms of the feedback toward icons, most participants indicated they provide very useful contextual information for them to perceive the meanings of the light colors more quickly ("I think the icon can connect to the state of the dog better."). And that two participants expected the size of the light box to be smaller to place in their desired locations. These feedback provide helpful design suggestions for future iteration of cAMpanion.

5 LIMITATIONS

Our design has four major limitations. First, cAMpanion currently displays 3 statuses, which may not include all behaviors owners are interested in seeing. A balance is needed to ensure users are not overwhelmed by too much information, losing the purpose of ambient display, while providing users the information they like to see. Second, cAMpanion currently displays information related to a dog's status without providing detailed information about the physical health status of dogs. Mapping these information together could provide users with more complete information, yet it might also cause information overload. Third, some participants mentioned the size of cAMpanion can be smaller. A balance between the visibility of light box (increase one's awareness via ambient display) and the space it occupied needs to be reached. Fourth, our initial user feedback suggested that users might expect more interactions through cAMpanion such as feeling the breathing of pet dogs or sending a reaction through device, as also indicated in prior research on ambient display [25]. Yet, the level of interactivity needs to be taken into account and reconsidered through further user studies, so the design will not lose the purpose and nature of ambient display.

6 CONCLUSION AND FUTURE WORK

In this paper, we presented the design and implementation of an ambient light box cAMpanion to promote distant owners' sense of connectedness to their dogs when living apart. The initial user feedback from our pilot study suggested that the icons on the light box along with different lightings allowed participants to completely capture their dogs' behaviors. However, the brightness of light, selection of light colors, as well as the interactivity of the breathing light can be reconsidered. As for the sensor deployment at the dog's side, it is important to consider the socket locations, as well as choose a specific sensor that can best capture the dog's particular behavior after taking into account its living environment and existing behavior patterns. For instance, we can obtain photos of dog's living environment from owners prior to sensor installation. If dogs are used to sleep or stay at multiple places in a home, multiple sensors can be installed to capture their status. And, the infrared sensor module can be used instead of the force sensor module if dogs do not sleep on any particular material (e.g., dog bed, sofa, human bed).

For future work, we will iterate the design of cAMpanion based on the initial user feedback. We will then deploy it in the field to capture participant reactions in a real-life context, and examine to

what extent *cAMpanion* can enhance the connectedness of distant owners to their dogs over time.

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